Rachel Cooray

Analyst, Data and AI, OCTAVE, John Keells Holdings

Colombo, Sri Lanka | rachelcooray@gmail.com | linkedin.com/in/rachel-cooray-069034235 | github.com/rachelcooray | ORCID 0009-0009-3192-2669 | rachelcooray.com

I apply AI to real business problems. Analyst, Data and AI at OCTAVE, John Keells Holdings, where I own four AI and analytics use cases for an FMCG business's distributors and design how each one is checked in the field. First Class Computer Science graduate from the University of Westminster, published researcher at IEEE ICAHS 2025, and co-founder of Layer1 Studio.

EXPERIENCE

Analyst, Data and AI, OCTAVE, John Keells Holdings (Colombo, hybrid) Jul 2026 to present

- Own delivery across four distributor-side use cases: Distributor Replenishment System (DRS), Ice Cream Distributor Margins (ICDM), Beverages Distributor Margins (BevDM), Permanent Dispatch Planning (PDP). Also contributed to Production Planning and weather-informed target setting.
- Coordinate data science and engineering teams to turn business requirements into implementable solutions, managing project plans, sprints, RAID logs, dependencies and priorities.
- Work directly with sales, finance, area and regional management, and present progress, findings and challenges to senior stakeholders.
- Write problem statements, process maps, decision logs and requirements, and support validation of data, model outputs, dashboards and reporting.
- Design measurement with pilot and control comparisons where possible and before and after analysis where appropriate, separating association from proven impact. Tools: Python, PySpark, SQL, Databricks, Delta Lake, Power BI, Excel.

Analytics Delivery Associate, OCTAVE, John Keells Holdings (Colombo) Apr 2026 to Jul 2026

Data Science and Analytics Delivery Intern, OCTAVE, John Keells Holdings (Colombo) Oct 2025 to Apr 2026

- Designed KPI-driven distributor incentive structures.
- Ran margin and sales-incentive simulations to recover lost sales in flood-affected outlets, and built Power BI dashboards for management.

Co-founder and Developer, Layer1 Studio (Colombo, Sri Lanka and London, UK, part-time) Dec 2025 to present

- I co-founded Layer1 Studio and build web and mobile products for small businesses, from shaping a brief to shipping and maintaining it.
- Work with clients to understand their needs and define project requirements. Develop websites and mobile applications. Contribute to interface design, application architecture and backend integration.
- Industries: fitness, marketing, arts and publishing, e-commerce, retail, food, workplace culture, sports. Mentor interns and involve them in real client projects.

Data Engineering Intern, London Success Academy (Remote, United Kingdom) Sep 2025 to Oct 2025

- Built ETL pipelines on GCP for clickstream and transaction data, scaled processing with PySpark and Parquet partitioning, and produced revenue, daily active user and conversion funnel insights.

Earlier internships at OCTAVE 2022 and 2023

Analytics Delivery Intern, Mar 2023 to May 2023. Data Science Intern, Mar 2022 to Jun 2022.

RESEARCH

PCOS Care: A Machine Learning-Based Web Application for Early Risk Prediction of Polycystic Ovary Syndrome, IEEE ICAHS 2025, Hammamet, Tunisia 2025

- DOI 10.1109/ICAHS66950.2025.11553077. With Malarvizhi Kaniappan Chinnathai and Salma Chahed, University of Westminster.
- An early screening web app with two levels of prediction: one from everyday signs that anyone can answer, and one that adds hormone test results. It shows which factors drove each prediction. It is a screening aid, not a diagnostic tool. Method: Chi-Square and ANOVA feature selection, SMOTE for class balance, and an enhanced Random Forest.
- Third place at the University of Westminster final year project showcase, 2025.
- Extended paper selected for the MSIT Journal special edition, under review.
- Deployed app: pcos-care-web-app.onrender.com. ORCID 0009-0009-3192-2669.

EDUCATION

BSc (Hons) Computer Science, First Class Honours, University of Westminster, London, United Kingdom — Sep 2022 to Jul 2025

Data Science specialisation. Started at IIT, Sri Lanka. UOW/IIT School of Computing and Engineering Part-Fee Scholarship, £5,000, 2024/25.

Certificate in Business Accounting, CIMA — Aug 2024

BA1 to BA4 completed. Operational level in progress.

St. Bridget's Convent, Colombo, GCE A/L 2022 and GCE O/L 2018 — 2008 to 2022

A/L: Chemistry A, English A, Physics B, Combined Mathematics B. O/L: nine A passes. Head Prefect, 2020.

BUILD LOG

Neurodiversity support chatbot — 2024

A chatbot prototype that guides people newly diagnosed with neurodiverse conditions, and their carers, through what comes next. Microsoft EMBRACE hackathon finalist. Generative AI, Conversational design, Prototyping.

Carbon Footprint Tracker

Turns everyday activities into an estimated carbon footprint using activity-based emission factors, with dashboards and suggestions for cutting it. Python, Flutter, Serverpod.

Futsal team recommender

A Flutter app that matches players to teams using collaborative and content-based filtering, with centralised stadium booking. Flutter, Node, Flask, Machine learning.

Fantasy Coin Collector

A 2D dungeon crawler built from scratch in HTML5 Canvas, with its own physics, procedural maps and sound. HTML5 Canvas, JavaScript, Web Audio API.

CSR street-vendor analytics — 2022

A field survey of street food vendors whose findings the CSR team adopted. Data collection, Data analytics, Field research.

SKILLS

Languages: Python, SQL, JavaScript, Dart

Data and AI: PySpark, Databricks, Power BI. Explored: scikit-learn, pandas, RAG pipelines

Web and mobile: React, Node.js, Flutter

Cloud and tools: GCP, Git and GitHub, Excel

TEACHING, MENTORING AND OTHER ROLES

Sunday School Class Teacher, St. Mary's Church, Dehiwala — Aug 2022 to Aug 2024

Student Support Worker, Clear Links Support — Oct 2024 to Apr 2025

Mentoring interns, Layer1 Studio — Dec 2025 to present

Customer-facing roles while studying in the UK

Customer Advisor, UNIQLO, Aug 2025 to Oct 2025. Customer Service Associate, EG On The Move, Jan 2025 to Sep 2025. Retail Associate, TK Maxx, Nov 2024 to Jan 2025.

RECOGNITION

Third place, final year project showcase, University of Westminster, judged with Netcompany and SmartDCC — 2025

Hackathon finalist, Generative AI for Neurodiversity, Microsoft EMBRACE Employee Resource Group — 2024

UOW/IIT School of Computing and Engineering scholarship, University of Westminster and Informatics Institute of Technology — 2024/25

School Prefect of the Year, New Generation Awards, Women in Management and the National Youth Services Council — 2020